

Name: _____ Date: _____ Period: _____

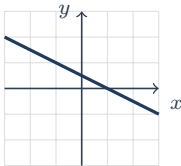
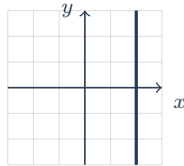
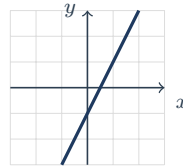
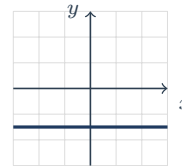
Instructions: Show all work. Write every line in $y = mx + b$ form when asked. No notes or calculator unless your teacher says so. **Total: 100 pts**

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers differ. Work every problem with no notes, then check your answers against the key.

Part A — Types of Slope (8 pts)

For each line, write whether its slope is **positive**, **negative**, **zero**, or **undefined**.

1. **negative**2. **undefined**3. **positive**4. **zero**

No graph this time — decide from the points or the equation.

5. The line through (3, 2) and (-1, 2): **zero**6. The line $y = 5x - 1$: **positive**7. The line $x = -4$: **undefined**8. The line through (0, 6) and (2, 1): **negative****Part B — Writing Equations of Lines (20 pts)**

Write each line in slope-intercept form, $y = mx + b$. Show how you found m and b ; for 11 and 12, start in point-slope form, $y - y_1 = m(x - x_1)$.

9. The line through (1, 5) and (3, 9).

$$m = \frac{9 - 5}{3 - 1} = \frac{4}{2} = 2$$

$$5 = 2(1) + b \Rightarrow b = 3$$

Equation: $y = 2x + 3$

10. The line through $(-4, 5)$ and $(2, -4)$.

$$m = \frac{-4 - 5}{2 - (-4)} = \frac{-9}{6} = -\frac{3}{2}$$

$$5 = -\frac{3}{2}(-4) + b = 6 + b \Rightarrow b = -1$$

Equation: $y = -\frac{3}{2}x - 1$

11. The line with slope 2 that passes through $(3, 4)$.

$$y - 4 = 2(x - 3)$$

$$y - 4 = 2x - 6$$

$$y = 2x - 2$$

Equation: $y = 2x - 2$

12. The line with slope $-\frac{1}{3}$ that passes through $(6, 1)$.

$$y - 1 = -\frac{1}{3}(x - 6)$$

$$y - 1 = -\frac{1}{3}x + 2$$

$$y = -\frac{1}{3}x + 3$$

Equation: $y = -\frac{1}{3}x + 3$

Part C — Piecewise Functions (20 pts)

13. **Evaluate.** (8 pts) $f(x) = \begin{cases} 3x + 2, & x < -1 \\ -1, & -1 \leq x \leq 1 \\ 2x - 5, & x > 1 \end{cases}$

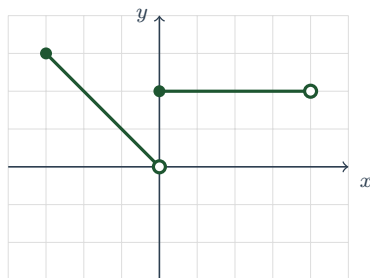
$f(-2) = \underline{-4}$ $f(-1) = \underline{-1}$ $f(1) = \underline{-1}$ $f(4) = \underline{3}$

14. **Domain.** (12 pts) Give the domain of each function as an inequality or in interval notation.

(a) $g(x) = \begin{cases} x - 2, & x < 3 \\ 7, & 3 \leq x < 6 \end{cases}$ Domain: $x < 6, (-\infty, 6)$

(b) $h(x) = \begin{cases} x + 1, & -2 \leq x \leq 0 \\ 4 - x, & 0 < x \leq 5 \end{cases}$ Domain: $-2 \leq x \leq 5, [-2, 5]$

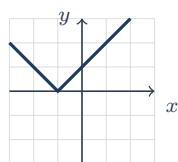
(c) The function k graphed below.



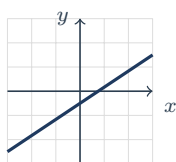
Domain: $-3 \leq x < 4$, $[-3, 4)$

Part D — The Absolute Value Function (28 pts)

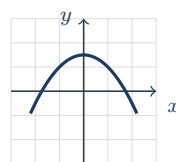
15. (3 pts) Circle the letter of the graph of an **absolute value** function.



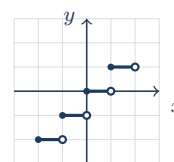
→ (A)



(B)



(C)



(D)

16. (3 pts) Which equation is an **absolute value** function?

(A) $y = x^2 - 3$

(B) $y = 2x + 5$

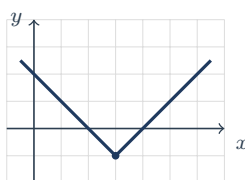
→ (C) $y = |x + 1| - 3$

(D) $y = \lfloor x \rfloor + 1$

17. (15 pts) Each function is a transformation of the parent $y = |x|$. Fill in the table.

Function	Vertex	Left/right shift	Up/down shift	Opens
$g(x) = x + 2 + 3$	<u>$(-2, 3)$</u>	<u>left 2</u>	<u>up 3</u>	<u>up</u>
$h(x) = x - 5 - 4$	<u>$(5, -4)$</u>	<u>right 5</u>	<u>down 4</u>	<u>up</u>
$k(x) = - x - 2$	<u>$(0, -2)$</u>	<u>none</u>	<u>down 2</u>	<u>down</u>

18. (4 pts) Which equation matches the graph?



(A) $y = |x + 3| - 1$

(B) $y = |x - 3| + 1$

→ (C) $y = |x - 3| - 1$

(D) $y = -|x - 3| - 1$

19. (3 pts) Compared with the parent $y = |x|$, the graph of $y = \frac{1}{2}|x|$ is

(A) narrower

→ (B) wider

(C) shifted down 2

(D) shifted left 2

Part E — Absolute Value Equations & Inequalities (24 pts)

Solve. Show both cases.

20. (8 pts) $|2x + 1| = 9$

$$\begin{aligned} 2x + 1 &= 9 & \text{or} & & 2x + 1 &= -9 \\ 2x &= 8 & \text{or} & & 2x &= -10 \end{aligned}$$

Solution: $x = 4$ or $x = -5$

21. (8 pts) $|x - 3| \leq 2$

$$\begin{aligned} -2 &\leq x - 3 \leq 2 \\ 1 &\leq x \leq 5 \end{aligned}$$

Solution: $1 \leq x \leq 5$

22. (8 pts) $|x + 4| > 6$

$$\begin{aligned} x + 4 &> 6 & \text{or} & & x + 4 &< -6 \\ x &> 2 & \text{or} & & x &< -10 \end{aligned}$$

Solution: $x > 2$ or $x < -10$